

NATIONAL INSTITUTE OF BUSINESS MANAGEMENT  
Diploma in Software Engineering (DSE20.2F/CO)

**COMPUTER TECHNOLOGY**  
27th October, 2021, 12:00 pm - 03:00 pm

**Answer all questions.**  
Calculators are allowed.

**(3 Hours)**

**1.**

a) Apply De Morgan's Theorems to the following expressions and **simplify**:

i.  $Z = \overline{(\overline{A + B + C}) \cdot (\overline{A + C})}$

ii.  $Z = \overline{(\overline{B} \cdot (A + \overline{B + C})) + (\overline{A} \cdot B \cdot \overline{C} + A \cdot \overline{C})}$

b)  $Z = F(A, B, C, D)$

$Z = 1$  for the minterms (1, 2, 3, 4, 7, 8, 12, 14)

$Z = \text{don't care}$  for the minterms (0, 9, 13)

$Z = 0$  for the remaining minterms

Simplify Z using Tabular Method.

**(20 marks)**

**2.**

a) Implement a Full Adder circuit using 2 half adders and an OR gate

b) A security system to detect an intruder to a warehouse has four sensors S1, S2, S3 and S4 connected to the house. The alarm(Y) connected to the system will be ON if the following conditions occur:

- If sensors S2 and S3 are activated at the same time or
- If sensors S2 and S4 are activated at the same time or
- If at least 3 of the four sensors are activated at the same time

i. Determine the truth table for the system

ii. Obtain the simplified SOP expression

iii. Implement the circuit using:

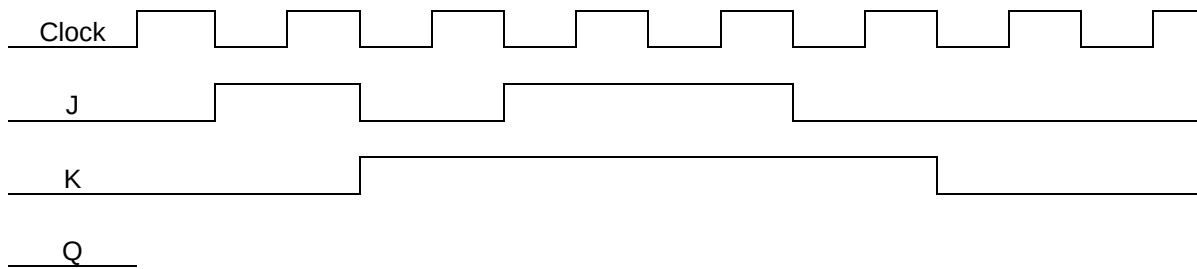
a. **NAND** gates only

b. A suitable Multiplexer.

c) Explain the operation of a **Demultiplexer**, taking a 1:4 DEMUX as an example.

**(20 marks)**

- 3.
- Explain the differences between sequential circuits and combinational circuits.
  - Write the reduced truth table of a JK flip-flop and derive its excitation table.
  - Complete the timing diagram for output of the positive edge triggered JK flip-flop. Assume that initial output of the flip-flop is logic low.



- Design a Sequential Logic Circuit to generate the count sequence of 1,2,3,4 using positive edge triggered JK flip-flops.
- Three edge-triggered JK flip flops in a synchronous circuit have the following input conditions:
 

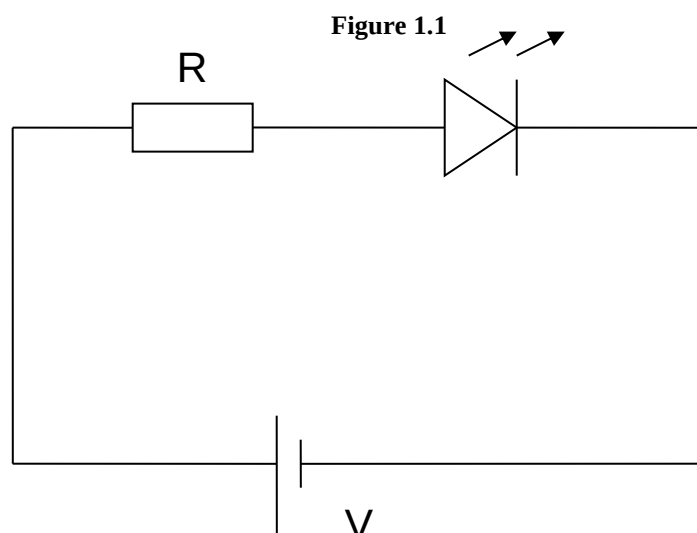
|             |             |
|-------------|-------------|
| $J_A = Q_B$ | $K_A = 1$   |
| $J_B = 1$   | $K_B = Q_A$ |
| $J_C = Q_B$ | $K_C = 1$   |

Assume that the initial state is  $Q_A = 1$   $Q_B = 1$   $Q_C = 1$ . Find the count sequence.

(20 marks)

4.

- Describe the construction and characteristics of two types of extrinsic semiconductors.
- Draw a full-wave rectifier circuit with capacitor smoothing and explain its operation using appropriate waveforms.
- A student plans to measure the current going through the Light Emitting Diode (LED) which presents in the circuit given by Figure 1.1. Figure 1.2 also depicts the characteristics curves of the LED. Copy the circuit in Figure 1.1 to your answer script and answer the following:



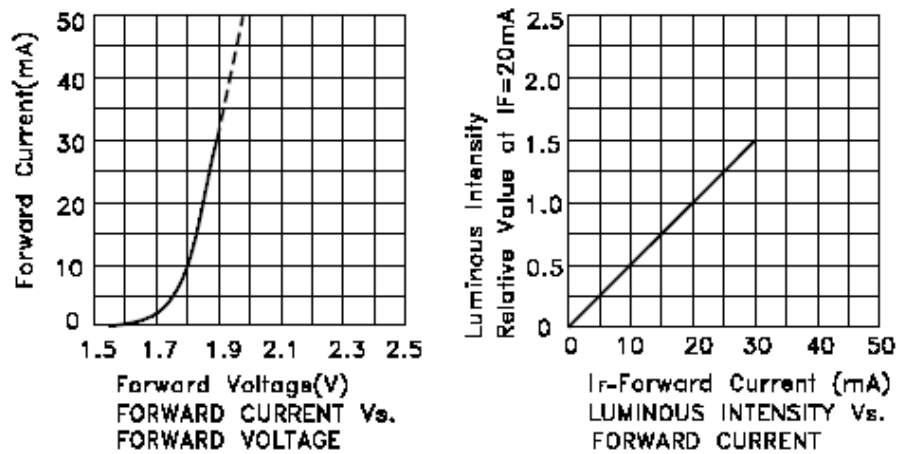


Figure 1.2

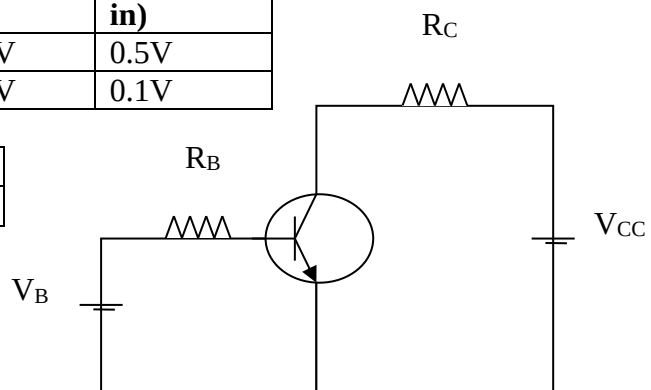
- What is the instrument used to measure the expected measurement?
- Explain briefly the method used to obtain the measurement and complete the circuit diagram including the instrument that you suggest.
- If the value obtained in the experiment is 10mA, what would be the Voltage across the LED and relative luminous intensity value?
- The resistance of the resistor R is 2.2 kilo ohms. Calculate the potential difference of the external potential V. **(20 marks)**

5.

- Draw IV characteristic curve for a diode identifying different regions
- If three resistors with a resistance of  $7\Omega$  is connected in series and in parallel what is the total resistance in each case.
- Consider the following circuit assume it to be in the active mode. Hence calculate  $I_B$ ,  $I_C$  and  $V_{CE}$ . Transistor is made out of Si.

| Material | $V_{CE}(\text{sat})$ | $V_{BE}(\text{sat})$ | $V_{BE}(\text{act})$ | $V_{BE}(\text{cut in})$ |
|----------|----------------------|----------------------|----------------------|-------------------------|
| Si       | 0.2V                 | 0.8V                 | 0.7V                 | 0.5V                    |
| Ge       | 0.1V                 | 0.3V                 | 0.2V                 | 0.1V                    |

| $\beta$ | $R_B$        | $R_C$       | $V_{CC}$ | $V_{BB}$ |
|---------|--------------|-------------|----------|----------|
| 30      | 90K $\Omega$ | 3K $\Omega$ | 12V      | 6V       |



- Explain the operation of an AND gate using the Transistor-Resistor logic

**(20 marks)**